

Rajesh Sudi

Embedded Systems Engineer | BSP | Device Drivers | Customer-Facing Support | Training & Mentoring
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Technical Leadership Summary

Field Applications Engineer with strong background in embedded systems, customer support, and hardware debugging. 15+ years of experience in troubleshooting, system integration, and supporting users in embedded platforms. Skilled in bridging customer requirements with technical solutions.

Silicon & Platform Expertise

- Board Bring-Up & Silicon Validation for ARM Cortex-based Platforms
- Embedded Linux BSP Development & Yocto-based Platform Customization
- Kernel Configuration, Driver Porting & Hardware Abstraction Layer Design
- Peripheral Enablement: I2C, SPI, UART, USB, Ethernet, PCIe
- Interrupt Handling, ISR Optimization & Deterministic Latency Control
- DMA Integration & Throughput Optimization
- Memory Subsystem Understanding – Heap/Stack Analysis & Mapping
- Cross-functional Debugging with Hardware & Validation Teams
- Customer Support & Technical Troubleshooting

Professional Experience

Senior Embedded Engineer – TenXer Labs (2023–Present)

- Leading BSP bring-up and firmware architecture for ARM-based embedded platforms.
- Developing interrupt-driven device drivers and validating peripheral interfaces.
- Collaborating with hardware teams during board initialization and silicon enablement.
- Optimizing real-time scheduling and memory usage in RTOS-based systems.
- Driving root-cause analysis for low-level hardware-software interaction issues.
- Design and development of **embedded firmware in Embedded C** for IoT, BLE, and Edge AI platforms.

Embedded Linux & Research Systems (2009–2023)

- Led embedded medical device and IoT prototype developments.
- Scientific Officer supporting PhD-level embedded research projects.
- Developed embedded Linux systems and driver integrations.
- Mentored engineering teams in embedded systems and AI integration.
- Supported semiconductor-aligned research in signal processing and RF MEMS.
- Architected scalable firmware layers for IoT and AI-enabled ARM systems.
- Mentored engineering teams on kernel internals and RTOS scheduling models.

Early Industry Foundation

- Project Manager – BRAINS (2005–2006): Unmanned navigation & communication systems.
- Testing & Process Engineer – Solectron, Wipro GE, Lucent, BEL (2001–2005).
- Electronics validation, hardware troubleshooting & reliability testing.

- Led unmanned ship navigation and underwater communication system development.
- Worked on embedded hardware integration and system-level debugging.
- Designed and developed a comprehensive fee management system tailored for educational institutions, enabling automated billing, payment tracking, and financial reporting using C#.
- Designed and developed an inventory management system to track and manage outgoing product dispatch details using C# and Visual Basic 6.0.

Education

- **M.Tech. Embedded System Design** – Jain University, 2019 (**Gold Medalist**).
- **M.Sc. (Engg.) by Research, ECE (RF MEMS)** – VTU, 2016.
- **B.E. Telecommunication Engineering** – VTU, 2009.
- **Diploma in Electronics & Communication** – DTE, 2001.

Certifications

- Python Programming – Coursera.
- Mathematics for Machine Learning – Coursera.
- IoT Development – Udemy.
- Jira Project Management – Coursera.

Achievements

- Best Project of the Year – KSCST
- IEEE Senior Member & Technical Committee Roles
- Mentor for Toycathon 2021 (Power Judging Round Selection)
- Reviewer – IEEE DISCOVER 2021, ICDSNS2025, IACIS2025.
- Authored and copyrighted a Toycathon idea.
- Technical expert committee member for setting up an IoT lab at MEI Polytechnic, Bengaluru.

Professional Memberships

- **IEEE Senior Member** (Branch Counsellor, IEEE JIT).
- **Executive Committee Member**, IEEE Signal Processing Society Bangalore.
- **Life Member**, ISTE.

Research & Publications

- Published papers in IEEE Conferences, IJERT, and reputed journals
- Areas: RF MEMS, IoT, Embedded Systems, Biomedical AI Applications.

DECLARATION

I hereby declare that the above information is true to the best of my knowledge.

Rajesh Sudi